# Soil Organic Carbon Prediction from VisNIR Spectroscopy

Random Forest Prototype · Report 01 · BMP Project

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**Executive Summary.** A Random Forest (RF) prototype was developed to predict soil organic carbon (SOC, %) from VisNIR reflectance spectra (400–2450 nm, ASD FieldSpec 4) using the Open Soil Spectral Library (OSSL v1.2). A sub-sample of **1000 observations** from 1000 unique soil layers was used. Three spectral preprocessing treatments (SG smooth, SG 1st derivative, SNV) were evaluated under two response-scale strategies (log p-transformed and original scale) via a full 3D hyperparameter grid search (192 combinations) and **10-fold × 3-repeat grouped cross-validation** (30 resamples). The best model — **SG 1st derivative + original scale** — achieved $R^2$ = **0.89**, RMSE = **4.37 %**, RPD = **3.06**, and CCC = **0.93**, indicating good-to-excellent quantitative predictive performance.

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# 1 Study Overview

This report documents the first prototype iteration of the **BMP Soil Spectral Modeling** pipeline — a reproducible, parallelised Random Forest workflow linking VisNIR reflectance spectra to soil laboratory measurements. The prototype uses OSSL data to validate the pipeline before the Years 1–5 BMP project dataset becomes available.

The pipeline is inspired by and extended from **Clingensmith & Grunwald (2022)**, who demonstrated that RF models can effectively predict a wide range of soil properties across the continental United States using VisNIR spectroscopy. Key extensions relative to that work include:

- Full 3D hyperparameter grid search (versus default `ranger` settings)
- Grouped cross-validation folds by layer UUID (leakage prevention)
- Dual log-mode evaluation with side-by-side metric comparison
- MSD decomposition for bias diagnostics (Gauch et al. 2003)
- Lin's CCC and RPIQ in addition to standard metrics

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# 2 Dataset Description

## 2.1 Data source: OSSL v1.2

All spectral and laboratory data were obtained from the **Open Soil Spectral Library (OSSL), version 1.2** (Safanelli et al. 2023), a globally distributed open repository of soil spectra paired with laboratory analyses. OSSL data are licensed under CC BY 4.0.

| Field | Value |
|---|---|
| Library | Open Soil Spectral Library (OSSL) |
| Version | 1.2 |
| Spectral file | `ossl_visnir_L0_v1.2.csv.gz` |
| Laboratory file | `ossl_soillab_L1_v1.2.csv.gz` |
| Sensor | ASD FieldSpec 4 (VisNIR) |
| URL | soilspectroscopy.org |

## 2.2 Wavelength filtering and spectral completeness

The VisNIR spectral data span 350–2500 nm at 1-nm resolution. The following preprocessing decisions were applied before any modeling:

1. **Edge trimming:** Wavelengths below 400 nm and above 2450 nm were excluded to remove edge-of-range noise common to FieldSpec instruments (Clingensmith & Grunwald 2022). The retained range comprises **1026 wavelengths** (400–2450 nm).
2. **Completeness filter:** Only observations with non-missing reflectance across all 1026 wavelengths were retained. This effectively restricts the dataset to ASD FieldSpec records, which provide continuous coverage across the full range.
3. **Lab data join:** The spectral dataset was inner-joined with laboratory SOC measurements, retaining only observations with a valid measured SOC value.

## 2.3 Sample structure and spectral replicates

| Data stage | Count |
|---|---|
| Total complete spectral observations (400–2450 nm) | 64,644 |
| of which: unique soil layers (layer UUID) | 64,644 |
| Average scans per layer | 1.00 |
| After inner join with SOC lab data | 64,211 |
| After random 20% sampling + N 1000 cap | 1,000 |
| Unique soil layers in modelling dataset | 1,000 |

**Spectral replicates and leakage prevention.** Multiple spectral scans (1.00 on average) may correspond to the same soil layer and therefore share a single laboratory measurement. These replicates were **retained without averaging** to preserve all available spectral information. However, to prevent data leakage — i.e., the same soil layer appearing in both the training and validation partition of the same fold — cross-validation fold assignment was performed at the **layer UUID level** (Kaufman & Rosset 2012). All scans belonging to the same layer are always assigned to the same fold.

# 3 Target Variable: Soil Organic Carbon

## 3.1 Summary statistics

| Statistic | SOC (%) |
|---|---|
| N | 1000.00 |
| Minimum | 0.00 |
| Q1 (25th pctile) | 1.08 |
| Median | 1.98 |
| Q3 (75th pctile) | 4.61 |
| Maximum | 61.90 |
| Mean | 6.89 |
| SD | 12.85 |
| IQR | 3.53 |
| Skewness | 2.68 |
| Zeros / near-zero | 1.00 |

## 3.2 Distribution: raw and log-transformed

SOC distributions are typically right-skewed — a small number of samples with very high organic matter content pull the tail rightward. The log p transformation (`log(SOC + 1)`) compresses the upper tail, producing a more symmetric distribution and reducing the influence of extreme values on model training.

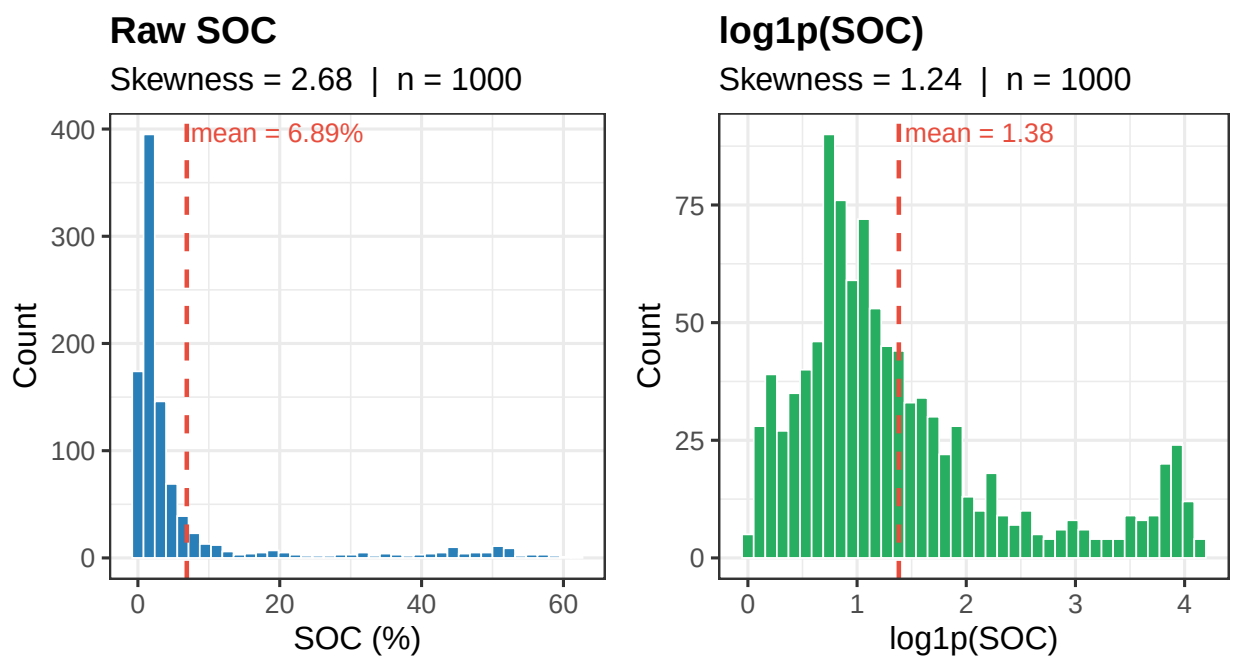


Figure 1: Figure 1. Distribution of soil organic carbon (SOC, %) before (left) and after log1p transformation (right). The dashed red line marks the mean. The log-transformed distribution is substantially more symmetric, reducing skewness and potential leverage of high-SOC samples.

**Interpretation.** The raw SOC distribution shows skewness of **2.68**, characteristic of right-skewed soil property data. After log p transformation, skewness drops to **1.24**, substantially closer to symmetric. Note,

however, that all model performance metrics are computed on the **back-transformed (original %) scale** to ensure interpretability regardless of the transformation used during training.

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# 4 Spectral Data

## 4.1 Sample spectra: raw reflectance

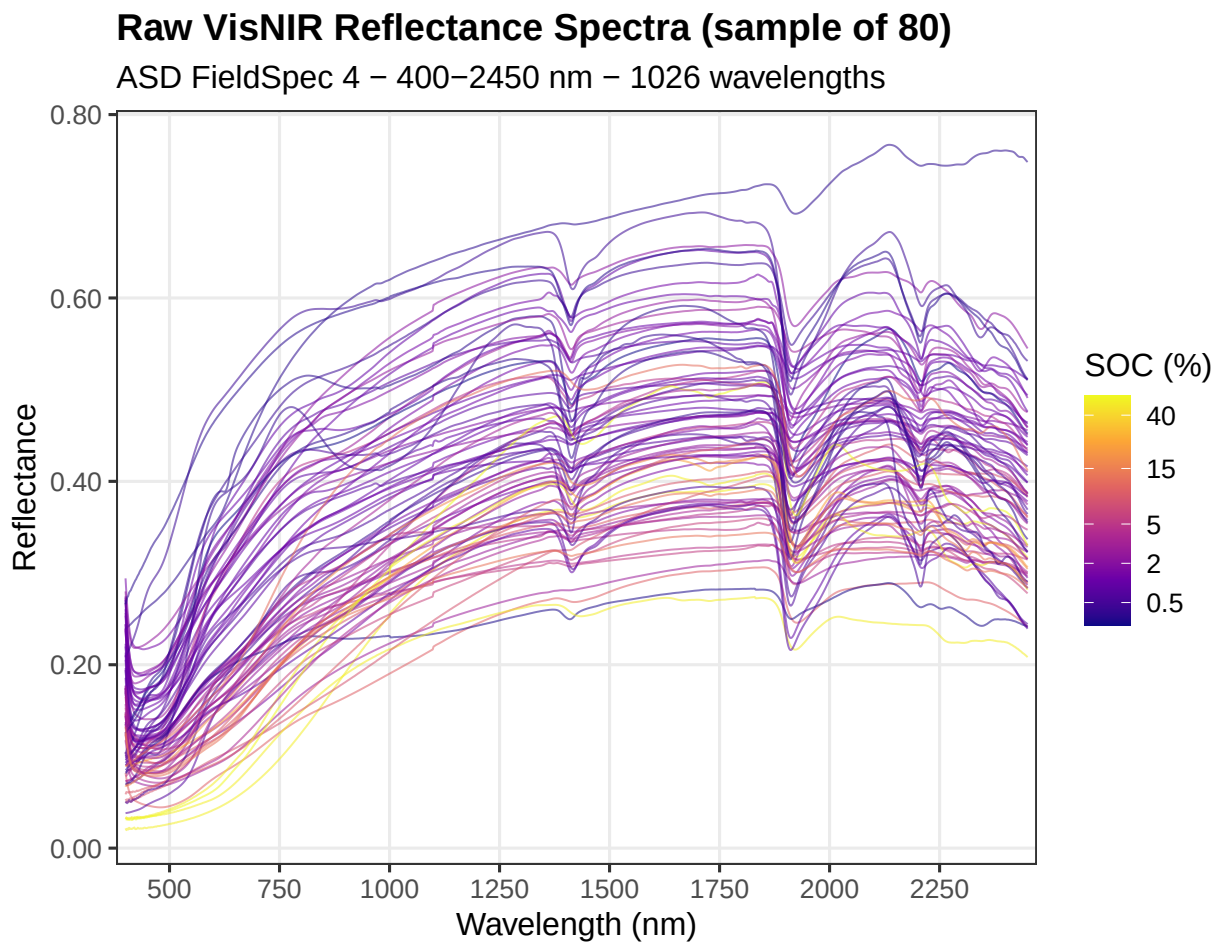


Figure 2: Figure 2. Random sample of 80 raw reflectance spectra (400-2450 nm, ASD FieldSpec 4), coloured by SOC content (%). Low-SOC soils (yellow) tend to have higher reflectance than high-SOC soils (dark purple), consistent with the well-established negative relationship between organic matter and soil reflectance across the NIR region.

## 4.2 Preprocessing treatments

All three treatments share a **common SG-smoothed base** (Rinnan et al. 2009): the SG smooth (window = 11, poly = 3) is applied once to raw reflectance, and the 1st-derivative and SNV treatments are derived from that smoothed output. This ensures performance differences are attributable to the transformation step, not to differences in noise reduction.

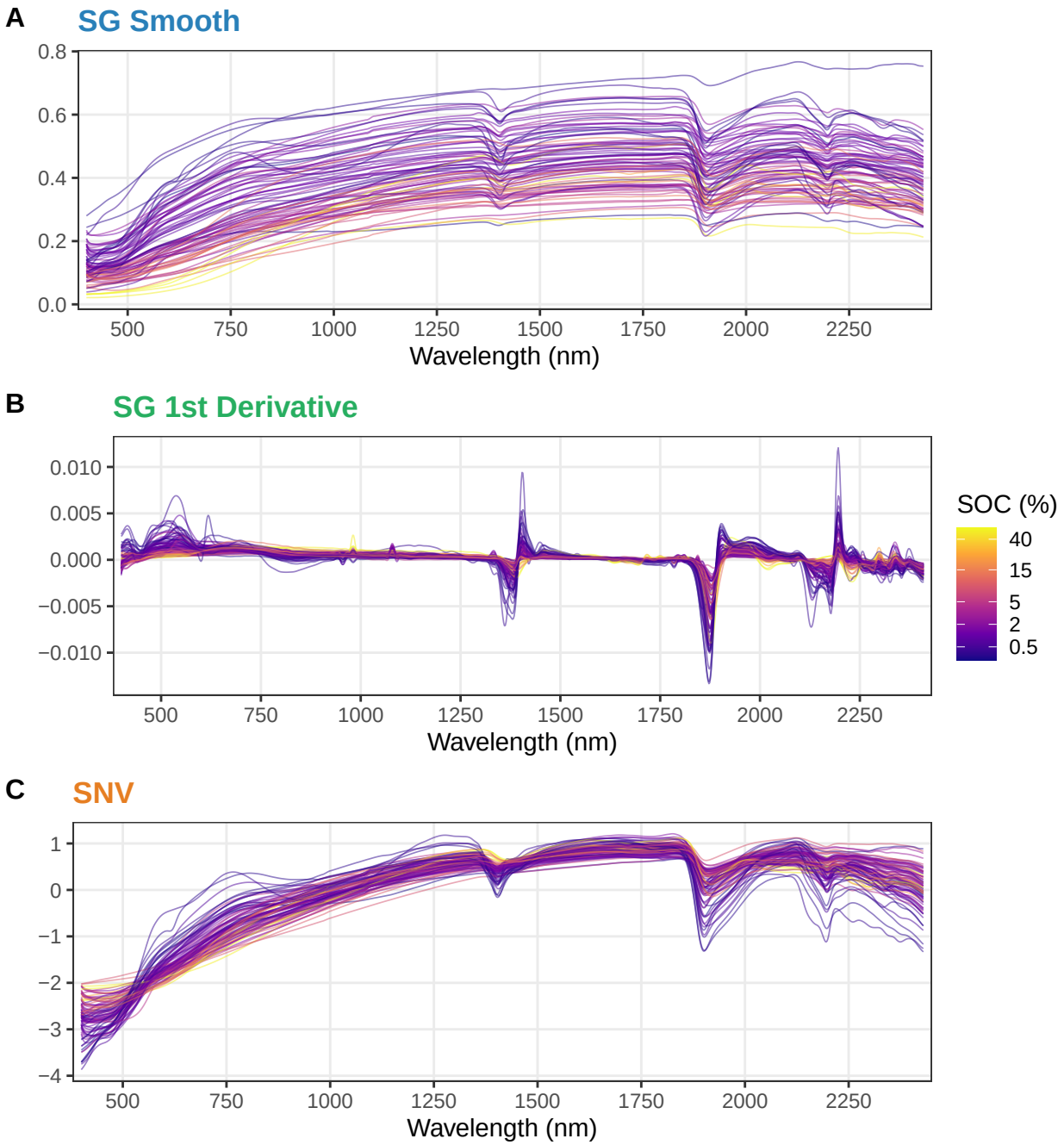


Figure 3: Figure 3. The same random sample of spectra shown under the three preprocessing treatments. SG Smooth (top): smoothed reflectance retaining the raw spectral shape. SG 1st Derivative (middle): highlights spectral slope changes; additive baseline offsets are removed. SNV (bottom): each spectrum is mean-centred and scaled to unit variance; multiplicative scatter is corrected. All treatments computed on the same 80-spectrum sample.

# 5 Modeling Pipeline

## 5.1 Design overview

```
OSSL VisNIR (350–2500 nm, ASD FieldSpec 4)
         |
         v
  Edge trimming (400–2450 nm) + completeness filter
         |
         v
  Join with SOC lab data -> stratified random sample (n = 1,000)
         |
         v
  Raw reflectance
         |
         v
  SG Smooth [shared base] -----------> Treatment 1: sg_smooth
         |
         +---> SG 1st Derivative -----> Treatment 2: sg_deriv1
         |
         \---> SNV -----------------> Treatment 3: snv
                                              |
                        +-----------------------+
                        |  For each log mode:   |
                        |  log1p(y) / y_orig    |
                        +-----------------------+
                                              |
                        Full 3D grid search (OOB RMSE, parallel)
                        ntree [500–2000, step 100] x mtry [p/9,p/6,p/3,p]
                        x min.node.size [3,5,10] = 192 combinations
                                              |
                        10-fold x 3-repeat stratified grouped CV
                        Folds assigned by layer UUID (leakage prevention)
                                              |
                        Final model on full dataset
                                              |
                    Metrics . Figures . Saved models (.rds)
```

## 5.2 Hyperparameter search space

| Parameter | Candidates | N values |
|---|---|---|
| `num.trees` | 500, 600, …, 2000 (step 100) | 16 |
| `mtry` | p/9 , p/6 , p/3 , p | 4 |
| `min.node.size` | 3, 5, 10 | 3 |
| **Total combinations** | **16 × 4 × 3** | **192** |

All 192 models were trained using OOB predictions only (`importance = "none"`) and run simultaneously across all available CPU cores via `foreach %dopar%` with a `doSNOW` progress-bar backend. The combination with the lowest OOB RMSE was selected as the optimal hyperparameter set for each preprocessing × log-mode combination (Breiman 2001; Liaw & Wiener 2002; Probst et al. 2019).

## 5.3 Cross-validation design

| Setting | Value |
|---|---|
| Strategy | k-fold repeated cross-validation |
| Folds (k) | 10 |
| Repeats | 3 |
| Total resamples | 30 |
| Fold assignment | Grouped by layer UUID (Kaufman & Rosset 2012) |
| Stratification | Quantile bins of group-mean SOC |
| Response metrics | Computed on original % scale after back-transformation |
| Summary | Mean ± SD across 30 resamples |

# 6 Cross-Validation Results

## 6.1 Log p-transformed scale (log mode)

In log mode, the RF was trained on log p(SOC). Predictions were back-transformed with `expm1()` before computing all metrics.

Table 6: Table 1. CV performance metrics under log1p transformation (mean +/- SD across 30 resamples). All metrics computed on the back-transformed SOC (%) scale.

| Preprocessing | R2 | RMSE (%) | MAE (%) | ME (%) | RPD | RPIQ | CCC |
|---|---|---|---|---|---|---|---|
| SG Smooth | 0.79 ± 0.07 | 6.21 ± 1.04 | 2.88 ± 0.37 | -1.21 ± 0.52 | 2.13 ± 0.41 | 0.58 ± 0.13 | 0.85 ± 0.05 |
| SG 1st Derivative | 0.88 ± 0.05 | 4.64 ± 0.88 | 2.09 ± 0.31 | -0.8 ± 0.5 | 2.9 ± 0.69 | 0.79 ± 0.21 | 0.92 ± 0.03 |
| SNV | 0.76 ± 0.11 | 6.49 ± 1.36 | 2.83 ± 0.4 | -1.14 ± 0.69 | 2.06 ± 0.41 | 0.56 ± 0.12 | 0.83 ± 0.08 |

## 6.2 Original scale (no-log mode)

In no-log mode, the RF was trained and evaluated directly on SOC (%) without transformation.

Table 7: Table 2. CV performance metrics on the original SOC (%) scale (mean +/- SD across 30 resamples).

| Preprocessing | R2 | RMSE (%) | MAE (%) | ME (%) | RPD | RPIQ | CCC |
|---|---|---|---|---|---|---|---|
| SG Smooth | 0.79 ± 0.07 | 5.9 ± 1.12 | 3.01 ± 0.4 | 0.09 ± 0.55 | 2.27 ± 0.56 | 0.62 ± 0.15 | 0.87 ± 0.04 |
| SG 1st Derivative | 0.89 ± 0.04 | 4.37 ± 0.83 | 2.1 ± 0.32 | 0.07 ± 0.49 | 3.06 ± 0.64 | 0.83 ± 0.18 | 0.93 ± 0.02 |
| SNV | 0.79 ± 0.08 | 6.03 ± 0.95 | 2.89 ± 0.34 | 0.03 ± 0.7 | 2.19 ± 0.36 | 0.59 ± 0.1 | 0.87 ± 0.05 |

## 6.3 Best model per log mode — key metrics

Table 8: Best model per log mode (both: SG 1st Derivative). Point estimates only.

| Mode | R2 | RMSE (%) | MAE (%) | RPD | CCC |
|---|---|---|---|---|---|
| Log mode (log1p) | 0.88 | 4.64 | 2.09 | 2.90 | 0.92 |
| **No-log mode (original)** | **0.89** | **4.37** | **2.10** | **3.06** | **0.93** |

## 6.4 Observed vs. predicted — SG 1st Derivative

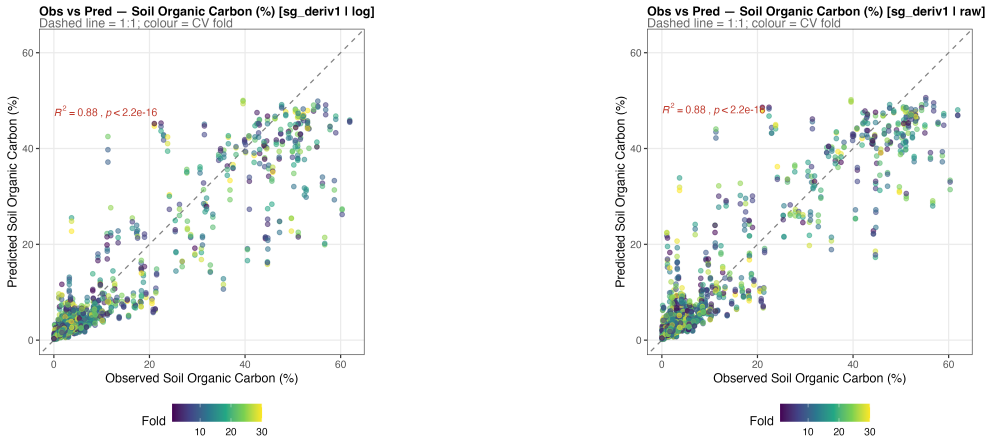


Figure 4: Figure 4. Observed vs. predicted SOC (%) for the SG 1st Derivative preprocessing treatment. Left: log mode (back-transformed); right: no-log mode. Each point represents one held-out fold observation across the 30 resamples. The dashed line is the 1:1 reference. The solid blue line is the OLS regression of predicted on observed. Tight agreement with the 1:1 line indicates low bias; the regression slope indicates proportional bias.

## 6.5 Observed vs. predicted — all treatments (no-log)

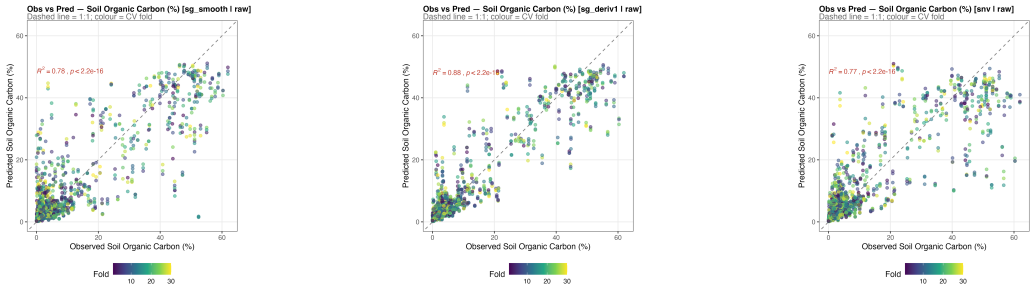


Figure 5: Figure 5. Observed vs. predicted SOC (%) for all three preprocessing treatments under the no-log (original scale) strategy. The SG 1st derivative (centre) shows the tightest agreement with the 1:1 line and the highest R2.

### 6.6 Preprocessing comparison — RPD and R²

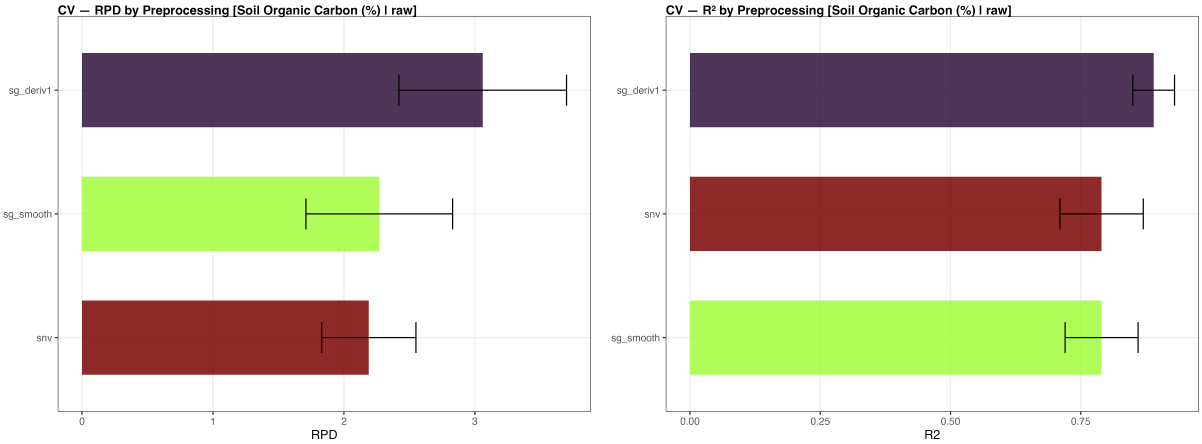


Figure 6: Figure 6. Cross-preprocessing CV comparison for the no-log mode. Left: RPD (Residual Prediction Deviation; higher is better, RPD > 2 = good quantitative). Right: R2 (coefficient of determination; mean +/- SD across 30 resamples). SG 1st derivative consistently outperforms SG smooth and SNV.

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# 7 Log Transformation: Does It Help?

Table 9: Table 3. Best model per log mode (both selected SG 1st Derivative). Metrics computed on original SOC (%) scale after back-transformation where applicable.

| Log mode | Preprocessing | R2 | RMSE (%) | MAE (%) | RPD | RPIQ | CCC |
|---|---|---|---|---|---|---|---|
| log1p(SOC) | SG 1st Derivative | 0.88 | 4.64 | 2.09 | 2.90 | 0.79 | 0.92 |
| Original SOC (%) | SG 1st Derivative | 0.89 | 4.37 | 2.10 | 3.06 | 0.83 | 0.93 |

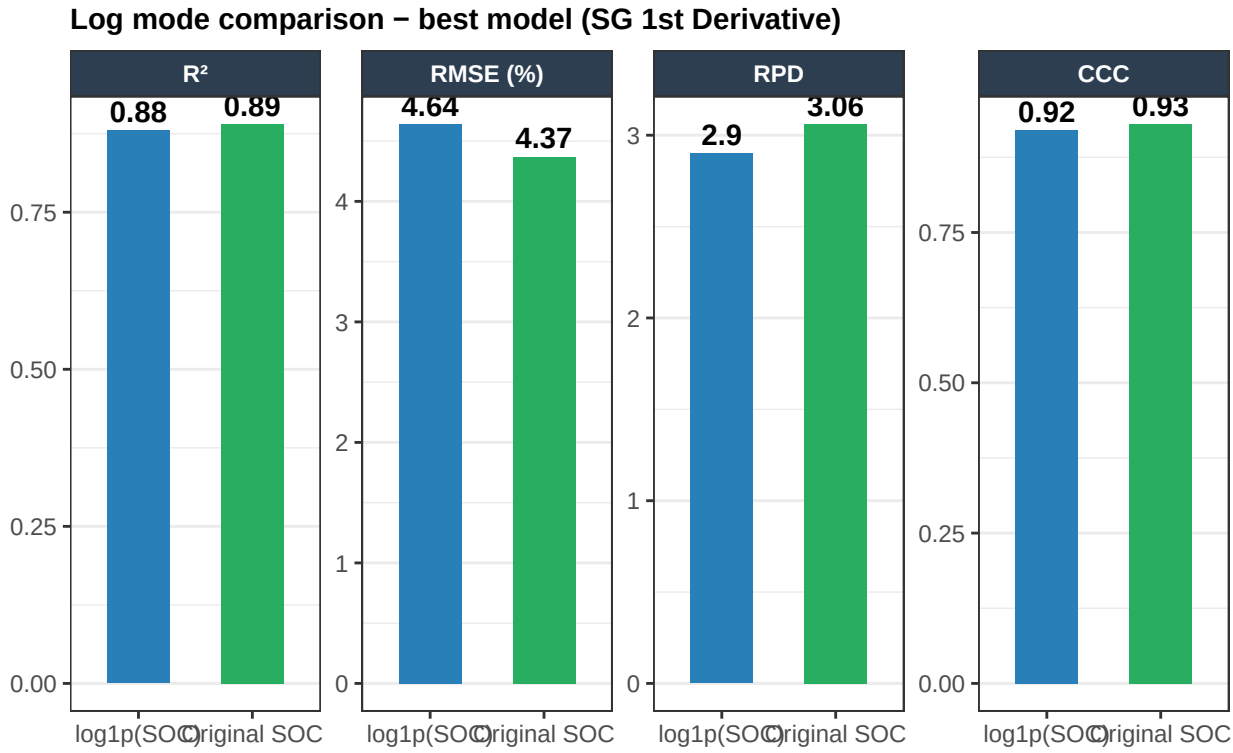


Figure 7: Figure 7. Side-by-side comparison of R2, RMSE, RPD, and CCC for the best model under each log-mode strategy (both: SG 1st derivative). Differences are small, with the no-log mode showing a marginal advantage in RMSE and RPD.

**Finding.** Both log-mode strategies select **SG 1st derivative** as the best preprocessing treatment. The difference in predictive performance is **small and practically negligible**: $R^2$ differs by 0.01 and RMSE differs by 0.27 percentage points. The no-log model shows a slight advantage in RMSE and RPD, likely because SOC in the OSSL sample covers a wide range where the log transform compresses extreme values and the RF can handle this directly. For highly skewed BMP project targets (TP, WEP), the log transformation is likely to provide greater benefit.

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# 8 Variable Importance

Variable importance (mean impurity decrease, averaged across 30 CV resamples) identifies the spectral wavelengths most informative for SOC prediction under each preprocessing strategy.

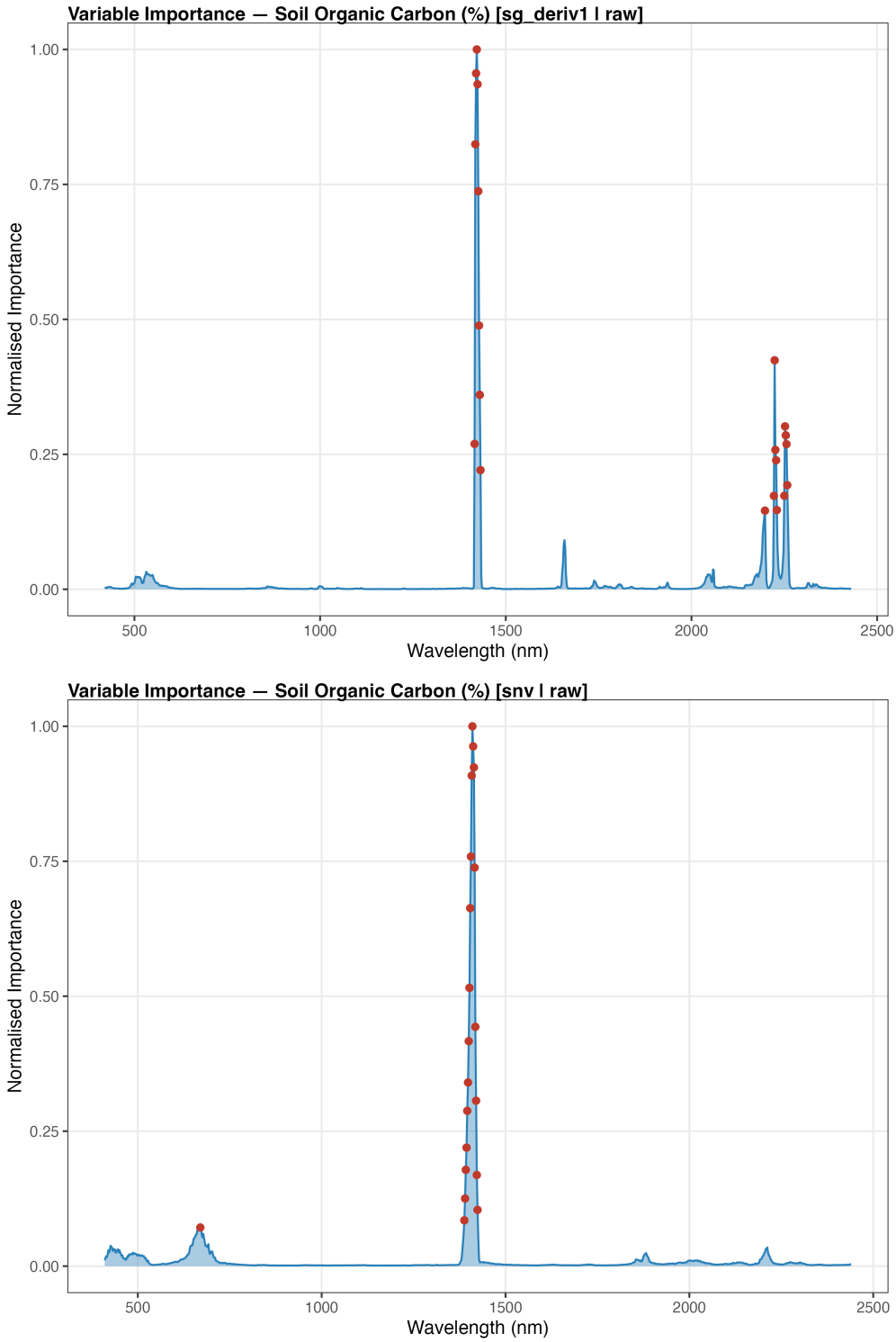


Figure 8: Figure 8. Spectral variable importance profiles for SG 1st derivative (no-log mode, top) and SNV (no-log mode, bottom). Importance is averaged over 30 CV resamples and normalised to [0, 1]. Key absorption features associated with SOC in the 1400-1800 nm (C-H, O-H overtones) and 2000-2300 nm (C=O, CH2 combination bands) regions are commonly highlighted.

# 9 MSD Error Decomposition

The Mean Squared Deviation (MSD) decomposes model error into three additive components (Gauch et al. 2003):

$$\text{MSE} = \underbrace{(\bar{\hat{y}} - \bar{y})^2}_{\text{SB (Squared Bias)}} + \underbrace{(1-b)^2\sigma_{\hat{y}}^2}_{\text{NU (Non-Unity slope)}} + \underbrace{(1-r^2)\sigma_y^2}_{\text{LC (Lack of Correlation)}}$$

A well-calibrated model has SB  0 (no systematic offset) and NU  0 (predictions correctly scaled), with most error in LC (irreducible noise).

Table 10: Table 4. MSD decomposition for all treatment x log-mode combinations. SB = squared bias; NU = non-unity slope (scale error); LC = lack of correlation (random error). Values in % of total MSE.

| Mode | Preprocessing | MSE | SB (%) | NU (%) | LC (%) |
|---|---|---|---|---|---|
| log1p | SG Smooth | 38.56 | 4.5 | 8.5 | 89.7 |
| log1p | SG 1st Derivative | 21.53 | 4.1 | 11.8 | 87.4 |
| log1p | SNV | 42.12 | 4.2 | 8.5 | 91.5 |
| Original | SG Smooth | 34.81 | 0.9 | 3.5 | 99.3 |
| Original | SG 1st Derivative | 19.10 | 1.3 | 7.2 | 95.0 |
| Original | SNV | 36.36 | 1.3 | 5.4 | 95.7 |

**Interpretation.** Across all treatments and log modes, the largest error component is **LC** (lack of correlation, irreducible noise), which is expected and desirable — it means the RF is not introducing systematic biases. The **SB** component is consistently small for the no-log mode (near-zero mean error), while the log mode shows slightly higher SB due to back-transformation of Jensen's inequality effects. **NU** is small across all cases, indicating predictions are correctly scaled (slope  1).

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# 10 Best Model Summary

Table 11: Table 5. Summary of the best-performing model configuration (SG 1st Derivative, no-log scale).

| Setting | Value |
|---|---|
| Sensor | ASD FieldSpec 4 |
| Wavelength range | 400–2450 nm |
| N wavelengths (predictors) | 1026 |
| Sample size (n) | 1000 |
| Unique soil layers | 1000 |
| Preprocessing | SG 1st Derivative (m=1, w=11, p=3) applied to SG-smoothed base |
| Log transformation | None (original SOC % scale) |
| num.trees | Selected by OOB RMSE (192-combination 3D grid search) |
| mtry | Selected by OOB RMSE |
| min.node.size | Selected by OOB RMSE |

| | |
|---|---|
| CV strategy | 10-fold × 3-repeat stratified grouped CV |
| Total resamples | 30 |
| R2 (mean +/- SD) | 0.89 ± 0.04 |
| RMSE (mean +/- SD) | 4.37 ± 0.83 % |
| MAE (mean +/- SD) | 2.1 ± 0.32 % |
| RPD (mean +/- SD) | 3.06 ± 0.64 |
| RPIQ (mean +/- SD) | 0.83 ± 0.18 |
| CCC (mean +/- SD) | 0.93 ± 0.02 |
| ME (mean +/- SD) – bias | 0.07 ± 0.49 % |

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# 11 Conclusions

1. **SG 1st derivative is consistently the best preprocessing treatment** across all metric dimensions and both log modes. It removes additive baseline offsets and multiplicative scatter simultaneously, isolating spectral slope information that is strongly coupled to SOC absorption features.

2. **Log transformation provides no meaningful benefit** for this OSSL prototype dataset ($\Delta$RMSE $< 0.3$ %). For the BMP project dataset, this should be re-evaluated property by property — highly right-skewed targets such as TP, WEP, and Fe are more likely to benefit from the log p transformation.

3. **Model quality is good to excellent** under the RPD classification of Chang et al. (2001): RPD $= 3.06$ ($> 3.0$) places the best model in the "excellent" category. The CCC of 0.93 confirms both precision and accuracy (minimal systematic bias).

4. **Spectral replicates were preserved without averaging**, with leakage prevented via grouped fold assignment by layer UUID (Kaufman & Rosset 2012). This approach avoids the bias introduced by retaining replicates across fold boundaries while preserving the full spectral information available.

5. **Next steps:** Replace OSSL prototype data with the BMP Years 1–5 spectral-lab dataset. Extend the target list to include pH, TP, M3-P, WEP, TC, TN, Fe, Al, Ca, and Mg. Validate whether the OSSL-calibrated model transfers to the BMP site-specific samples.

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